Dean Athony Pitts **PROG7312**

ST10449407 **POE Part 1 Task 1**

Youtube: https://youtu.be/ndCsIYxb7mU

Github: https://github.com/EMKNDN/emkndn-prog7312-g2-2026-prog7312-poe-deano708/tree/main

User Engagement Strategy 1:

Real time visual telemetry feedback is the instant response of a system to an input or change in variables. This is a highly relevant user engagement strategy allowing users of a system to instantly notice a change in variables or environment on their system. This can be seen in the example of a similar racing system using real time feedback. “Dashboards help bring all that data together, and the lap comparison relays performance in real-time, enabling analysts to compare the different laps second-by-second. This helps to determine what factors may have impacted any slow-down in performance, or even what might have contributed to an exceptional, record-beating lap (Burnett-Thompson, 2023).” This is a relevant example as this ties in closely to a trucking system such as Smart X Trucks which need to monitor onboard hardware sensors and have quick feedback upon an environmental change.

User Engagement Strategy 2:

Proactive alert gamification is the concept of creating interactive and rewarding notification components aimed at improving user engagement with a system. Some studies mention “There is clear benefit in designing and sending notifications to users that persuade them to interact with an app and marketer goals (Kunkel, Hayduk and Lock, 2023).” Although not typical for an enterprise system such as Smart X Trucks, having a gamified and proactive alert system would boost user engagement and ensure users are fully aware of any essential changes that would be otherwise ignored if portrayed as a typical notification component.

User Engagement Strategy 3:

Simplified onboarding flows is the process of reducing user cognitive load by introducing an intuitive and educational experience when the user first logs into the system while keeping this process simple. This ensures users are engaged by quickly gaining an overview of the system.” The onboarding process sets the tone for the customer's relationship with your brand. By providing customers with the knowledge they need, businesses can empower them to use their products more effectively, leading to higher satisfaction and reduced churn rates (Komandla, 2019).” This is possible for a trucking enterprise system such as Smart X Trucks which

would highly benefit from users gaining an understanding of the system through the use of a simplistic and educational onboarding UI.

User Engagement Strategy 4:

Progressive disclosure is an “HCI principle that involves gradually revealing information to users as needed rather than all at once (Muralidhar, Belloum and Ashok, 2025).” This principle introduces a hierarchy of information allowing users to focus on primary components and secondary components later. This is a relevant user engagement strategy for a system like Smart X Trucks as it’s easy for users to get distracted by many of the different sensors which is why a primary component and a secondary component showing sensors would be a sensible decision to make.

User Engagement Strategy 5:

Command Line Interface & Keyboard shortcuts. This user engagement strategy appeals to the professionals and developers of the interface world who prefer to use hotkeys and shortcuts for navigation instead of UI components. “Keyboard shortcuts are such a valuable resource for expert users that HCI practitioners have long recommended to support their use. Many blog posts from end-users further demonstrate the benefits of keyboard shortcuts to interact with operating system (e.g., Windows, applications (e.g., Photoshop, Office or Chatbots (e.g., ChatGPT) and increase productivity or reduce the strain caused by repetitive motions (Bailly et al., 2025).” Implementing shortcuts and command line functionality into Smart X Trucks would allow regular clients to quickly navigate and optimise their overall HCI experience without taking their hands off of their keyboard.

Justification of Choice:

I believe for a well put together logistics management dashboard such as Smart X Trucks, the best choice for user engagement would be real time visual telemetry feedback. As previously mentioned, real time visual telemetry feedback is the instant to almost instant response of a system to an input or change in variables.

Due to the Smart X Truck’s incorporation of multiple sensors, having this real time feedback with an intuitive visual telemetry design is a highly relevant choice allowing for users to stay engaged with sensor changes allowing the users to notice the smallest change in sensor data. Portraying this intuitive visual telemetry can be accomplished through many design choices:

Fleet map:

Shown on the live fleet dashboard page is the incorporation of a map showing the latest location presented by the GPS units assigned to each truck. This allows users of the dashboard to have a current knowledge of the location of the truck at one glance. Although only presented with fake data at the moment, this map allows the

backtracking of truck locations in hourly steps to a total of twelve hours. This interface component is perfect for real time visual feedback as it allows admin users to have a quick glance at where the truck is currently situated and where it has made progress over the last twelve hours.

Visual Design Elements:

The dashboard incorporates multiple visual elements allowing for an enhanced visual feedback experience. One of these visual elements is with the simple inclusion of colour coding on telemetry readings. As simple as this is, it plays a key role in quick visual recognition when scanning over multiple sensor readings of a large collection of trucks as seen on the dashboard page. Visual weighting also plays a key role in quick visual identification and feedback as having a clear hierarchy of sensor data reduces cognitive load by creating a clear direct of high priority to low priority such as oil level being of high importance and GPS transponder being of low importance.

References:

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Komandla, V. (2019). Effective onboarding and engagement of new customers: Personalized strategies for success. *SSRN Electronic Journal*, 8(12). doi:10.2139/ssrn.4983100.

Kunkel, T., Hayduk, T. and Lock, D. (2023). Push it real good: The effects of push notifications promoting motivational affordances on consumer behavior in a gamified mobile app. *European Journal of Marketing*, [online] 57(9). doi:10.1108/ejm-06-2021-0388.

Muralidhar, D., Belloum, R. and Ashok, A. (2025). Operationalizing selective transparency using progressive disclosure in artificial intelligence clinical diagnosis systems. *International Journal of Human-Computer Studies*, [online] 204, p.103591. doi:10.1016/j.ijhcs.2025.103591.

Declaration of AI Usage

Section	Purpose	AI Tool	Date	Link
TruckTelemetry.cs	Setup of TruckTelemetry including validation logic	Gemini	12/09/2026	https://share.gemini.google/O9Qf6fnMtm4B
KZNDataSeeder.cs	Creation of truck data records and seeding logic	Gemini	12/09/2026	https://share.gemini.google/O9Qf6fnMtm4B
TelemetryServices.cs	Setup of TelemetryBatchProcessor	Gemini	12/09/2026	https://share.gemini.google/O9Qf6fnMtm4B
MediaController.cs	Setup of API MediaController including upload and delete functionality	Gemini	12/09/2026	https://share.gemini.google/O9Qf6fnMtm4B
Dashboard.razor	Setup of initial dashboard	Gemini	14/09/2026	https://share.gemini.google/O9Qf6fnMtm4B
TruckSensorManagement.razor	Setup of TruckSensorManagement page	Gemini	14/09/2026	https://share.gemini.google/O9Qf6fnMtm4B
TruckSensorStateService.cs	Setup of TruckSensorStateService	Gemini	12/09/2026	https://share.gemini.google/O9Qf6fnMtm4B